Sathira Silva

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Interests

3D Perception | Vision-Language Models | Self-supervised Learning

Education

University of Peradeniya, Sri Lanka **Nov. 2018 – Dec. 2023**
B.Sc. (Hons) in Computer Engineering *GPA: 3.60 / 4.00*
[Courses, Thesis] *Class: 2nd Upper*

De Mazenod College **Jan. 2003 – August 2016**
G.C.E. Advanced Level *Maths - A, Physics - A, Chemistry - B*

Experience

Research Engineer **May 2025 – Current**
Computer Vision Department — Principal Investigator: *Dr. Muhammad Haris Khan* *MBZUAI*

- Developed **microCLIP**, an unsupervised CLIP adaptation method using saliency-guided token selection and coarse-fine token fusion for fine-grained image classification (Findings of ACL 2026).
- Led **BabyMind**, investigating object-first inductive biases for vision-language learning from child-view videos, with a focus on grounding and developmentally inspired representation learning (CogSci 2026).
- Contributed to **MedCache**, a training-free spatially aware caching framework for faster medical video generation (NeurIPS 2026).
- Made substantial contributions to the successful **SafeMaize** research proposal funded by the Gates Foundation through IAAI, particularly its technical research plan and methodology.
- Working on a CVPR-targeted SafeMaize study of domain shift and reliable maize screening, alongside field-data collection planning, agronomic annotation, and expert validation with project partners.

Research Assistant **June 2024 – April 2025**
Computer Vision Department — Principal Investigator: *Dr. Muhammad Haris Khan* *MBZUAI*

- Contributed to DPA, an unsupervised adaptation method for CLIP using dual visual and textual prototypes (WACV 2025).
- Developed fine-grained visual-textual interaction and multi-crop self-training pipelines for adapting pre-trained CLIP without target annotations.
- Investigated self-trained alignment scores for fine-grained CLIP adaptation, contributing to the method published at WACV 2026 (oral presentation).

Computer Vision Research Engineering Intern **Dec. 2022 – Mar. 2023**
Autonomous Vehicle R&D Division *Vega Innovations*

- Contributed to the integration of a transformer architecture called NEAT into an autonomous vehicle system, by reviewing the paper and understanding its internals.
- Developed real-time computer vision solutions for autonomous vehicles on high performance GPU inference embedded systems (Nvidia DRIVE PX2 / Jetson TX2).

Teaching Assistant: Programming Methodology (CO222) **May 2021 – Sep. 2021**
Department of Computer Engineering *University of Peradeniya*

- Supervised weekly 2hr long online lab sessions.
- Created questions for online quizzes based on the C programming Language.
- One-on-one sessions with students to tutor them on the C programming language concepts.

Selected Achievements

ACES Coders v7.0 | An inter-university algorithmic programming competition organized by of UoP **2020**
Rank - 14 / 100+
Team Name: bitLasagna

IEEEXtreme 14.0 | 24-hour global algorithmic programming competition **2020**
Country Rank - 2, Global Rank - 68 / 7000+
Team Name: InterGreat

ICDS Mini Hackathon | An inter-university Data Science Hackathon **2021**
Rank - 5 / 100+
Team Name: bitLasagna

IEEEXtreme 16.0 | 24-hour global algorithmic programming competition **2022**
Country Rank - 27, Global Rank - 427 / 6373
Team Name: bitLasagna

ACES Coders v9.0 | An inter-university algorithmic programming competition organized by of UoP **2022**
Rank - 2 / 100+
Team Name: bitLasagna

Publications

microCLIP: Unsupervised CLIP Adaptation via Coarse-Fine Token Fusion for Fine-Grained Image Classification **2026**

In Findings of the Association for Computational Linguistics (ACL 2026), pp. 33277–33294

Authors: Sathira Silva, Eman Ali, Chetan Arora, Muhammad Haris Khan

MedCache: Training-Free Spatially Aware Caching for Accelerated Medical Video Generation **2026**

In 40th Conference on Neural Information Processing Systems (NeurIPS 2026)

Authors: Ufaq Khan, Umair Nawaz, Sathira Silva, Numan Saeed, Muhammad Umer Sheikh, Muhammad Bilal, Junaid Qadir, Lena Maier-Hein, Yutong Xie, Muhammad Haris Khan

Learning Visual Referents from Noisy Child-view Videos **2026**

In Proceedings of ACM International Conference on Multimedia (ACM MM 2026)

Authors: Abrham Kahsay Gebreselasie, Sathira Silva, Michael Sidorov, Daniel Harari, Muhammad Haris Khan

Objects Before Words: Object-First Inductive Biases for Grounding Language in Child-View Video **2026**

In Proceedings of the 48th Annual Meeting of the Cognitive Science Society (CogSci 2026)

Authors: Sathira Silva, Abrham Kahsay Gebreselasie, Muhammad Umer Sheikh, Kartik Kuckreja, Daniel Harari, Muhammad Haris Khan

Towards Fine-Grained Adaptation of CLIP via a Self-Trained Alignment Score **2026**

[*Oral*] In 2026 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), pp. 5875–5885

Authors: Eman Ali, Sathira Silva, Chetan Arora, Muhammad Haris Khan

DPA: Dual Prototypes Alignment for Unsupervised Adaptation of Vision-Language Models **2025**
poster

In 2025 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), pp. 6083–6093

Authors: Eman Ali, Sathira Silva, Muhammad Haris Khan

S2TPVFormer: A Spatiotemporal Approach to Tri-Perspective Representation for 3D Semantic Occupancy Prediction **2025**
poster

[*Best Paper Award*] In AAAI 2025 Machine Learning for Autonomous Driving (ML4AD) Workshop

Authors: Sathira Silva, Savindu Wannigama*, Gihan Jayatilake, Muhammad Haris Khan, Roshan Ragel (*Equal contribution)*

Ongoing Projects

SafeMaize: Reliable Maize Screening under Domain Shift **2026 – Present**

Index Terms: Robust Visual Recognition, Domain Generalization, Selective Prediction *Research*

Technologies: PyTorch, CLIP / Vision-Language Models, Distributed Training

- Developing a CVPR-targeted research study and the **MaizeShift-RBench** benchmark to evaluate cross-source recognition, calibration, and risk-coverage under field-image domain shifts.
- Investigating symptom-aware visual representations and reliability when acquisition conditions obscure diagnostic evidence or unfamiliar symptoms appear.
- Contributing to the broader **Gates Foundation-funded project (via IAAI)**, including field-data collection protocols, Telugu farmer descriptions, agronomic annotation and expert review, and coordination with agricultural partners.

Technical Skills

Languages: C/C++, Python, Java, HTML/CSS, JavaScript, SQL

Developer Tools: Visual Studio, VS Code, Eclipse, Jupyter Notebook, Android Studio

Technologies/Frameworks: OpenMMLab, PyTorch, TensorFlow, Bash Scripting, GitHub, OpenCV, Transformers, CLIP, Distributed GPU Training, ReactJS, NodeJS, Jekyll

Certifications

Natural Language Processing (hons) **Jan. 2022**
HSE University *Coursera*

Algorithms on Graphs **July 2020**
University of California San Diego *Coursera*

Data Structures **June 2020**
University of California San Diego *Coursera*

Convolutional Neural Networks **Feb. 2020**
DeepLearning.AI *Coursera*

Neural Networks and Deep Learning **Jan. 2020**
DeepLearning.AI *Coursera*

Relevant Coursework

Data Structures & Algorithms
Operating Systems
Software Methodology
Computer Architecture
Image Processing
Programming Methodology
Artificial Intelligence
Discrete Mathematics
Networking and Web Application Design
Probability and Statistics

References

Dr. Muhammad Haris Khan

Assistant Professor,
Department of Computer Vision,
Mohammed bin Zayed University of AI,
United Arab Emirates

Prof. Roshan G. Ragel

Head of Department,
Department of Computer Engineering,
Faculty of Engineering,
University of Peradeniya, Sri Lanka